

COMPLETE FINANCIAL MATHEMATICS

Upper Secondary / Terminale Coursebook

Percentages • Interest • Annuities • Loans • Investments

30 Complete Lessons • Practice Activities
Revision Tests • Answer Key • Formula Reference

A structured 80-page learning book for classroom teaching, independent study and exam preparation.

FINANCIAL MATHEMATICS SERIES

Course focus	Practical quantitative finance, time value of money, consumer finance and investment mathematics.
Level	Upper Secondary / Terminale / introductory college-preparation.
Language	English.

GETTING STARTED

How to Use This Book

A practical roadmap for mastering financial mathematics step by step.

What You Will Learn

- Core percentage skills, simple and compound interest, present value and discounting.
- Annuities, loans, amortization, sinking funds and depreciation.
- Investment returns, NPV, IRR, bonds, shares and foreign exchange.
- Consumer finance, break-even analysis, inflation and risk.

Recommended Study Method

1. Read the objective and identify the main formula.
2. Copy the worked example and explain every symbol.
3. Complete guided practice without checking answers.
4. Correct errors by writing the rule behind the mistake.
5. Review difficult lessons after 24 hours and one week.

Financial Mathematics Checklist

- Percentage base: What amount is the percentage applied to?
- Rate: Is it decimal, annual, nominal, effective or periodic?
- Time: Are years, months and payment periods consistent?
- Timeline: When does each cash flow occur?
- Formula: Does it match PV, FV, annuity, loan or percentage problem?
- Answer: Is the sign, currency, percentage and size reasonable?

Book Organization

- Lessons 1-8: Percentages, interest, compounding, discounting and equations of value.
- Lessons 9-16: Annuities, perpetuities, loans, amortization and sinking funds.
- Lessons 17-24: Depreciation, inflation, returns, NPV, IRR, bonds and shares.
- Lessons 25-30: FX, payroll, markup, break-even, consumer finance, risk and planning.
- Pages 65-72: Eight review tests; pages 73-78: answers and skills; pages 79-80: formula references.

Suggested 6-Week Schedule

- Week 1: Lessons 1-5.
- Week 2: Lessons 6-10.
- Week 3: Lessons 11-15.
- Week 4: Lessons 16-20.
- Week 5: Lessons 21-25.
- Week 6: Lessons 26-30 + revision tests.

Lessons 1-15

Learning Map

- ## Study Notes

REFERENCE

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Learning Map

- Identify the formula family before calculating.
- Match rate period with cash-flow period.
- Use timelines for time-value-of-money problems.
- Interpret every numerical answer in its financial context.

Study Notes

LESSON 01 - CORE FINANCIAL MATHEMATICS

Percentages and Financial Ratios

Concepts, formulas and worked examples

Learning Objectives

- Convert between fractions, decimals and percentages.
- Calculate percentage increase and decrease.
- Use ratios to compare financial quantities.

Concept / Formula	Explanation	Worked Example
Percentage	part / whole $\times 100$	45 out of 60 = $45/60 \times 100 = 75\%$.
Percentage increase	$(\text{new} - \text{old}) / \text{old} \times 100$	Price rises from 80 to 92: increase = $12/80 \times 100 = 15\%$.
Percentage decrease	$(\text{old} - \text{new}) / \text{old} \times 100$	Price falls from 500 to 425: decrease = $75/500 \times 100 = 15\%$.
Ratio	Comparison of two quantities in the same or compatible units.	Profit : sales = 30 : 200 = 3 : 20.

Key Rules

- Always divide the change by the original value.
- A multiplier of $1+r$ gives an increase of r ; a multiplier of $1-r$ gives a decrease.
- Ratios should be simplified and interpreted in context.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. 15% of 240 = 36.
2. Increasing 400 by 8% gives 432.
3. A 20% discount means paying 80% of the original price.
4. A margin of 25% means profit is 25% of sales, not cost.

Problem-Solving Routine

Step 1: Identify known values, unknown value and timing.

Step 2: Convert rates and time units so periods match.

Step 3: Choose the formula and substitute carefully.

Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 01 - PRACTICE

Percentages and Financial Ratios

Common errors, guided practice and mastery check

Common error	Better method	Why?
Dividing percentage change by the new value.	Divide by the original value.	Percentage change is measured relative to the starting amount.
Adding 10% then subtracting 10% returns to the original.	Use multipliers: $1.10 \times 0.90 = 0.99$.	Successive percentage changes are multiplicative.

Guided Practice

- Find 18% of 350.
- Increase 640 by 12%.
- Reduce 900 by 25%.
- A price rises from 120 to 138. Find the percentage increase.
- Convert 0.375 to a percentage.
- Convert 62% to a decimal.
- Simplify the ratio 84:126.
- A profit is 45 on sales of 300. Find the profit margin.

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

- 63.
- 716.8.
- 675.
- 15%.
- 37.5%.
- 0.62.
- 2:3.
- 15%.

Independent Practice

- Write three new numerical examples based on this lesson.
- Change one given rate or time period and predict how the answer changes.
- Write one deliberately incorrect calculation, then correct it and explain the error.
- Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 02 - CORE FINANCIAL MATHEMATICS

Simple Interest

Concepts, formulas and worked examples

Learning Objectives

- Use the simple interest formula.
- Distinguish principal, rate, time and interest.
- Calculate maturity value.

Concept / Formula	Explanation	Worked Example
Simple interest	$I = P r t$	$P=1000$, $r=6\%=0.06$, $t=3$ gives $I=180$.
Maturity value	$A = P + I = P(1+rt)$	1000 at 6% simple for 3 years gives $A=1180$.
Rate conversion	Use annual decimal rate with time in years.	9 months = $9/12 = 0.75$ year.
Solve for rate	$r = I/(Pt)$	If $I=150$, $P=1000$, $t=2$, $r=7.5\%$.

Key Rules

- Simple interest is calculated only on the original principal.
- Convert percentage rates to decimals before substitution.
- Use consistent time units.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. 2000 at 5% for 4 years earns 400 interest.
2. 1500 at 8% for 6 months earns 60.
3. A loan of 5000 repaid as 5600 after 2 years has simple rate 6% p.a.
4. Simple interest grows linearly with time.

Problem-Solving Routine

- Step 1: Identify known values, unknown value and timing.
- Step 2: Convert rates and time units so periods match.
- Step 3: Choose the formula and substitute carefully.
- Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 02 - PRACTICE

Simple Interest

Common errors, guided practice and mastery check

Common error	Better method	Why?
Using $t=6$ for six months with an annual rate.	Use $t=6/12=0.5$ years.	Time must match the rate period.
Adding the rate directly to money.	Use $I=Prt$.	The rate is a proportion of principal per time period.

Guided Practice

- Find simple interest on 2400 at 7% for 3 years.
- Find maturity value of 5000 at 4.5% for 2 years.
- Find interest on 1800 at 10% for 9 months.
- Find the rate if 3000 earns 540 in 3 years.
- Find time if 2500 at 8% earns 600.
- Find principal if interest is 420 at 7% for 2 years.
- Does simple interest compound?
- What is the graph shape of amount against time?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

- 504.
- 5450.
- 135.
- 6% p.a.
- 3 years.
- 3000.
- No.
- A straight line.

Independent Practice

- Write three new numerical examples based on this lesson.
- Change one given rate or time period and predict how the answer changes.
- Write one deliberately incorrect calculation, then correct it and explain the error.
- Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 03 - CORE FINANCIAL MATHEMATICS

Compound Interest

Concepts, formulas and worked examples

Learning Objectives

- Use the compound amount formula.
- Compare compound and simple growth.
- Solve routine compound interest problems.

Concept / Formula	Explanation	Worked Example
Compound amount	$A = P(1+r)^n$	1000 at 5% for 3 years gives 1157.625.
Compound interest	$I = A - P$	Interest above = 157.625.
Growth factor	$1+r$	At 8%, annual growth factor is 1.08.
Decay factor	$1-r$	At 12% annual depreciation, factor is 0.88.

Key Rules

- Compound interest is calculated on accumulated balance.
- n counts compounding periods, not automatically years.
- Round money at the end unless instructions say otherwise.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. 2000 at 6% for 2 years becomes 2247.20.
2. 5000 at 4% for 5 years uses $5000(1.04)^5$.
3. Compound growth is exponential.
4. Small rate differences matter strongly over long periods.

Problem-Solving Routine

- Step 1: Identify known values, unknown value and timing.
- Step 2: Convert rates and time units so periods match.
- Step 3: Choose the formula and substitute carefully.
- Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 03 - PRACTICE

Compound Interest

Common errors, guided practice and mastery check

Common error	Better method	Why?
Using $A=P(1+nr)$.	Use $A=P(1+r)^n$.	Compound growth multiplies each period.
Rounding every intermediate year.	Keep calculator precision and round final money.	Early rounding can accumulate error.

Guided Practice

- Find amount on 1500 at 8% for 4 years.
- Find compound interest on 4000 at 5% for 3 years.
- Find amount on 12000 at 3% for 10 years.
- What is the growth factor for 6.5%?
- What is the decay factor for 9%?
- Compare simple and compound interest conceptually.
- What happens to interest each year under compounding?
- Why is compounding powerful over long periods?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

- 2040.73 approximately.
- 630.50 approximately.
- 16127.00 approximately.
- 1.065.
- 0.91.
- Simple uses original principal; compound uses accumulated balance.
- It generally increases when the balance grows.
- Interest earns interest.

Independent Practice

- Write three new numerical examples based on this lesson.
- Change one given rate or time period and predict how the answer changes.
- Write one deliberately incorrect calculation, then correct it and explain the error.
- Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 04 - CORE FINANCIAL MATHEMATICS

Compounding Frequency

Concepts, formulas and worked examples

Learning Objectives

- Use nominal rates compounded several times per year.
- Calculate periodic rate and number of periods.
- Compare annual, semiannual, quarterly and monthly compounding.

Concept / Formula	Explanation	Worked Example
General formula	$A = P(1+j/m)^{mt}$	j is nominal annual rate, m compounding periods per year.
Periodic rate	$i = j/m$	12% nominal compounded monthly gives 1% per month.
Number of periods	$n = mt$	3 years monthly gives $n=36$.
More frequent compounding	For same nominal j, more frequent compounding slightly increases effective yield.	Monthly usually exceeds annual at same nominal rate.

Key Rules

- Match rate period and number of periods.
- Do not divide an effective annual rate by m unless it is stated as nominal.
- Use $m=2,4,12$ for semiannual, quarterly, monthly.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. 1000 at 12% nominal monthly for 1 year = $1000(1.01)^{12}$.
2. Quarterly 8% nominal means 2% per quarter.
3. Five years semiannual means 10 periods.
4. Daily compounding uses a daily periodic rate under the stated convention.

Problem-Solving Routine

Step 1: Identify known values, unknown value and timing.

Step 2: Convert rates and time units so periods match.

Step 3: Choose the formula and substitute carefully.

Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 04 - PRACTICE

Compounding Frequency

Common errors, guided practice and mastery check

Common error	Better method	Why?
Using 12% each month.	Use $12\%/12 = 1\%$ monthly.	Nominal annual rate is split among periods.
Using $n=3$ for three years monthly.	Use $n=36$.	n counts compounding periods.

Guided Practice

- Find monthly rate for 9.6% nominal compounded monthly.
- Find quarterly rate for 8% nominal quarterly.
- How many monthly periods in 4 years?
- Find amount of 5000 at 6% nominal monthly for 2 years.
- Find amount of 3000 at 8% nominal quarterly for 3 years.
- Which gives more at same nominal rate: annual or monthly?
- What is m for semiannual compounding?
- What is n for 2.5 years quarterly?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

- 0.8% per month.
- 2% per quarter.
- 48.
- 5635.80 approximately.
- 3804.73 approximately.
- Monthly.
- 2.
- 10.

Independent Practice

- Write three new numerical examples based on this lesson.
- Change one given rate or time period and predict how the answer changes.
- Write one deliberately incorrect calculation, then correct it and explain the error.
- Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 05 - CORE FINANCIAL MATHEMATICS

Effective and Nominal Interest Rates

Concepts, formulas and worked examples

Learning Objectives

- Distinguish nominal and effective annual rates.
- Convert nominal compounded rates to effective annual rates.
- Compare borrowing or investment rates fairly.

Concept / Formula	Explanation	Worked Example
Effective annual rate	$EAR = (1+j/m)^m - 1$	12% nominal monthly gives $(1.01)^{12} - 1 = 12.6825\%$.
Nominal rate	Quoted annual rate divided across compounding periods.	$j=8\%$ quarterly means 2% per quarter.
Equivalent rates	Rates producing the same accumulation over the same time.	An EAR can be compared directly across products.
Periodic rate from EAR	$i_{\text{period}} = (1+EAR)^{(1/m)} - 1$	Convert annual effective to monthly equivalent.

Key Rules

- Use effective rates to compare products with different compounding frequencies.
- Nominal rates are not directly comparable unless compounding frequency is the same.
- Keep rates in decimal form in formulas.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. 6% nominal monthly has EAR above 6%.
2. An EAR of 10% means one-year growth factor 1.10.
3. Equivalent monthly rate to 12% EAR is $(1.12)^{(1/12)} - 1$.
4. Frequent compounding raises EAR for a fixed positive nominal rate.

Problem-Solving Routine

- Step 1: Identify known values, unknown value and timing.
- Step 2: Convert rates and time units so periods match.
- Step 3: Choose the formula and substitute carefully.
- Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 05 - PRACTICE

Effective and Nominal Interest Rates

Common errors, guided practice and mastery check

Common error	Better method	Why?
Calling 12% nominal monthly an EAR of 12%.	Calculate the effective rate.	Monthly compounding makes the one-year yield larger than 12%.
Comparing nominal 8% monthly with effective 8.1% without conversion.	Put both on an effective annual basis.	Comparison requires equivalent periods.

Guided Practice

- Find EAR for 12% nominal monthly.
- Find EAR for 8% nominal quarterly.
- Find one-year growth factor for EAR 7.5%.
- Why is EAR useful?
- Is nominal 10% monthly equivalent to 10% effective?
- Find periodic quarterly rate for 12% nominal quarterly.
- State formula for EAR.
- Which is larger at positive rates: nominal j or corresponding EAR with $m > 1$?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

- 12.6825% approximately.
- 8.2432% approximately.
- 1.075.
- It allows fair annual comparison.
- No.
- 3% per quarter.
- $(1+j/m)^m - 1$.
- EAR.

Independent Practice

- Write three new numerical examples based on this lesson.
- Change one given rate or time period and predict how the answer changes.
- Write one deliberately incorrect calculation, then correct it and explain the error.
- Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 06 - CORE FINANCIAL MATHEMATICS

Present Value and Discounting

Concepts, formulas and worked examples

Learning Objectives

- Calculate present value of future cash flows.
- Interpret discounting as reverse compounding.
- Use present value in financial decisions.

Concept / Formula	Explanation	Worked Example
Present value	$PV = FV/(1+r)^n$	1100 due in 1 year at 10% has $PV=1000$.
Discount factor	$1/(1+r)^n$	At 8% for 2 years: factor = $1/(1.08)^2$.
Time value of money	Money today can earn returns, so equal future money is worth less today.	PV converts future cash to present equivalent.
Multiple cash flows	Discount each cash flow to time 0 and add.	$PV = \sum CF_t/(1+r)^t$.

Key Rules

- Higher discount rates reduce present value.
- Longer waiting times reduce present value for positive rates.
- Discount each cash flow according to its timing.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. PV of 1210 in 2 years at 10% is 1000.
2. PV of a future payment supports loan and investment pricing.
3. A discount factor is always below 1 for positive r and positive n .
4. Cash flows at time 0 are not discounted.

Problem-Solving Routine

Step 1: Identify known values, unknown value and timing.

Step 2: Convert rates and time units so periods match.

Step 3: Choose the formula and substitute carefully.

Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 06 - PRACTICE

Present Value and Discounting

Common errors, guided practice and mastery check

Common error	Better method	Why?
Multiplying FV by $(1+r)^n$.	Divide by $(1+r)^n$.	Discounting reverses compounding.
Discounting all cash flows by the same exponent.	Use each cash flow time t .	Timing matters.

Guided Practice

- Find PV of 1331 due in 3 years at 10%.
- Find PV of 5000 due in 5 years at 6%.
- What happens to PV if r rises?
- What happens to PV if n rises?
- Find discount factor at 8% for 1 year.
- Why is money today valuable?
- How do you value multiple future cash flows?
- What is PV of a cash flow already at time 0?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

- 1000.
- 3736.29 approximately.
- It falls.
- It falls for positive rates.
- 0.9259 approximately.
- It can be invested or used now.
- Discount each to present and add.
- Its current amount; no discounting.

Independent Practice

- Write three new numerical examples based on this lesson.
- Change one given rate or time period and predict how the answer changes.
- Write one deliberately incorrect calculation, then correct it and explain the error.
- Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 07 - CORE FINANCIAL MATHEMATICS

Simple Discount and Bank Discount

Concepts, formulas and worked examples

Learning Objectives

- Understand simple discounting.
- Calculate proceeds under bank discount.
- Distinguish discount rate from simple interest rate.

Concept / Formula	Explanation	Worked Example
Bank discount	$D = F d t$	Face value F discounted at rate d for time t .
Proceeds	$P = F - D = F(1 - dt)$	Amount received now.
Face value	Amount due at maturity.	A note may have face value 10000.
Discount rate	Applied to face value, not proceeds.	This differs from simple interest on principal.

Key Rules

- Check whether a problem uses true interest or bank discount.
- Use time in years if d is annual.
- The effective yield on proceeds is usually greater than the quoted discount rate.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. $F=5000$, $d=6\%$, $t=0.5$ gives $D=150$ and proceeds= 4850 .
2. Discounting occurs before maturity.
3. Face value is repaid at maturity.
4. Bank discount uses F as the base.

Problem-Solving Routine

- Step 1: Identify known values, unknown value and timing.
- Step 2: Convert rates and time units so periods match.
- Step 3: Choose the formula and substitute carefully.
- Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 07 - PRACTICE

Simple Discount and Bank Discount

Common errors, guided practice and mastery check

Common error	Better method	Why?
Using proceeds as the base for D.	Use face value F in $D = Fdt$.	Bank discount is quoted on maturity value.
Confusing discount with interest earned on invested principal.	Identify the convention first.	The bases differ.

Guided Practice

- Find D on $F=8000$ at 7% for 9 months.
- Find proceeds for the same note.
- Define face value.
- What base is used in bank discount?
- Find D on 12000 at 5% for 4 months.
- What happens to proceeds if d rises?
- Why can bank discount yield exceed d on proceeds?
- State proceeds formula.

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

- 420.
- 7580.
- Amount due at maturity.
- Face value.
- 200.
- Proceeds fall.
- Because discount is charged on face value while proceeds are smaller.
- $P = F(1 - dt)$.

Independent Practice

- Write three new numerical examples based on this lesson.
- Change one given rate or time period and predict how the answer changes.
- Write one deliberately incorrect calculation, then correct it and explain the error.
- Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 08 - CORE FINANCIAL MATHEMATICS

Equivalent Rates and Equation of Value

Concepts, formulas and worked examples

Learning Objectives

- Move cash flows to a common focal date.
- Use an equation of value to compare obligations.
- Solve simple replacement-payment problems.

Concept / Formula	Explanation	Worked Example
Focal date	Chosen date to compare all cash flows.	Often time 0 or a settlement date.
Equation of value	Sum of values of one set at focal date equals sum of the other set.	Discount future payments or accumulate earlier ones.
Accumulation	Move cash forward: $V_{\text{future}} = V_{\text{now}}(1+r)^n$.	Use when focal date is later.
Discounting	Move cash backward: $V_{\text{now}} = V_{\text{future}}/(1+r)^n$.	Use when focal date is earlier.

Key Rules

- All cash flows in an equation of value must be expressed at the same date.
- Choose a focal date that simplifies the algebra.
- Use the stated interest convention consistently.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. A payment today and one in 2 years can be replaced by one payment in 1 year.
2. Move each amount to year 1 before equating.
3. Equivalent obligations have equal value at the chosen rate.
4. Changing the interest rate changes equivalence.

Problem-Solving Routine

Step 1: Identify known values, unknown value and timing.

Step 2: Convert rates and time units so periods match.

Step 3: Choose the formula and substitute carefully.

Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 08 - PRACTICE

Equivalent Rates and Equation of Value

Common errors, guided practice and mastery check

Common error	Better method	Why?
Adding amounts from different dates directly.	Move them to a common focal date.	Money values at different times are not directly comparable.
Using different rates for different sides without instruction.	Use the same stated valuation rate.	The equation of value needs a consistent basis.

Guided Practice

1. What is a focal date?
2. Why move cash flows to one date?
3. How do you move cash forward one year?
4. How do you move cash backward one year?
5. Can 100 today and 100 next year be added directly for valuation?
6. What does an equation of value express?
7. Does equivalence depend on the interest rate?
8. Give one common use of equation of value.

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

1. A chosen date for comparing cash flows.
2. To make values comparable.
3. Multiply by $1+r$.
4. Divide by $1+r$.
5. Not without first valuing them at a common date.
6. Equality of financial value at one date.
7. Yes.
8. Replacing several debts with one settlement payment.

Independent Practice

- A. Write three new numerical examples based on this lesson.
- B. Change one given rate or time period and predict how the answer changes.
- C. Write one deliberately incorrect calculation, then correct it and explain the error.
- D. Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 09 - CORE FINANCIAL MATHEMATICS

Annuities: Basic Concepts

Concepts, formulas and worked examples

Learning Objectives

- Define an annuity and payment period.
- Distinguish ordinary annuity and annuity due.
- Recognize annuity timelines.

Concept / Formula	Explanation	Worked Example
Annuity	Series of equal payments at regular intervals.	Monthly deposits of 100.
Ordinary annuity	Payments at end of each period.	Loan payments commonly modeled this way.
Annuity due	Payments at beginning of each period.	Rent may be paid at start of month.
Payment R	Constant periodic cash flow.	R=250 each month.

Key Rules

- Timeline position matters: beginning vs end changes value.
- The periodic interest rate must match the payment period.
- n is the number of payments/periods.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. Monthly payments require a monthly effective periodic rate.
2. An annuity due has each payment invested one period longer than an ordinary annuity.
3. Equal payments simplify closed-form formulas.
4. Unequal cash flows require individual discounting or other methods.

Problem-Solving Routine

- Step 1: Identify known values, unknown value and timing.
- Step 2: Convert rates and time units so periods match.
- Step 3: Choose the formula and substitute carefully.
- Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 09 - PRACTICE

Annuities: Basic Concepts

Common errors, guided practice and mastery check

Common error	Better method	Why?
Using annual rate directly with monthly payments.	Convert to matching monthly periodic rate.	Rate period must match payment period.
Ignoring first payment timing.	Identify ordinary or due.	Timing changes both present and future values.

Guided Practice

1. Define an annuity.
2. What is an ordinary annuity?
3. What is an annuity due?
4. Where is the first payment of an annuity due?
5. What must match the payment period?
6. What does R represent?
7. If payments are monthly for 5 years, how many payments?
8. Which generally has larger FV: same ordinary annuity or annuity due?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

1. Equal regular payments over time.
2. Payments at end of periods.
3. Payments at beginning of periods.
4. At time 0/start of first period.
5. The periodic interest rate.
6. Periodic payment.
7. 60.
8. Annuity due.

Independent Practice

- A. Write three new numerical examples based on this lesson.
- B. Change one given rate or time period and predict how the answer changes.
- C. Write one deliberately incorrect calculation, then correct it and explain the error.
- D. Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 10 - CORE FINANCIAL MATHEMATICS

Future Value of an Ordinary Annuity

Concepts, formulas and worked examples

Learning Objectives

- Calculate future value of end-of-period deposits.
- Interpret the annuity accumulation factor.
- Solve for periodic payment.

Concept / Formula	Explanation	Worked Example
FV ordinary annuity	$FV = R \left[\frac{(1+i)^n - 1}{i} \right]$	Deposit R each period at rate i.
Accumulation factor	$s\text{-angle-}n = \frac{(1+i)^n - 1}{i}$	Future value per unit payment.
Solve for R	$R = FV / \text{factor}$	Used for savings targets.
No interest case	$FV = Rn$ when $i=0$	Simple sum of payments.

Key Rules

- Use periodic rate i and number of payments n .
- Ordinary annuity deposits occur at period ends.
- The last payment earns no interest before valuation at the same instant.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. 100 monthly for 12 months at 1% monthly: $FV=100[(1.01^{12}-1)/0.01]$.
2. A savings plan target can be converted to required monthly deposit.
3. Higher i increases FV.
4. More payments increase FV.

Problem-Solving Routine

- Step 1: Identify known values, unknown value and timing.
- Step 2: Convert rates and time units so periods match.
- Step 3: Choose the formula and substitute carefully.
- Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 10 - PRACTICE

Future Value of an Ordinary Annuity

Common errors, guided practice and mastery check

Common error	Better method	Why?
Multiplying R by $(1+i)^n$ only once.	Use the annuity factor.	Each payment compounds for a different length of time.
Treating annuity due as ordinary.	Multiply ordinary FV by $1+i$ for an annuity due.	Each due payment earns one extra period.

Guided Practice

1. State FV ordinary annuity formula.
2. Find factor for $i=0$ if $n=20$.
3. What does R represent?
4. What happens to FV when i rises?
5. What happens to FV when n rises?
6. Why does each payment not earn the same interest?
7. How do you solve for R from target FV?
8. Does the final ordinary payment earn one full period before valuation?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

1. $FV = R \left[\frac{(1+i)^n - 1}{i} \right]$.
2. 20.
3. Periodic payment.
4. FV rises.
5. FV rises.
6. Payments are made at different times.
7. Divide target FV by the annuity factor.
8. No.

Independent Practice

- A. Write three new numerical examples based on this lesson.
- B. Change one given rate or time period and predict how the answer changes.
- C. Write one deliberately incorrect calculation, then correct it and explain the error.
- D. Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 11 - CORE FINANCIAL MATHEMATICS

Present Value of an Ordinary Annuity

Concepts, formulas and worked examples

Learning Objectives

- Calculate present value of equal end-of-period payments.
- Interpret annuity present-value factor.
- Use PV to value loans and payment streams.

Concept / Formula	Explanation	Worked Example
PV ordinary annuity	$PV = R[1 - (1+i)^{-n}]/i$	Present value of n end-period payments.
PV factor	$a\text{-angle-}n = [1 - (1+i)^{-n}]/i$	PV per unit payment.
Solve for payment	$R = PV / \text{factor}$	Loan installment calculation.
Limiting behavior	Higher i lowers PV of same future payments.	Discounting is stronger.

Key Rules

- Payments are discounted back to time 0.
- Rate i must match payment period.
- PV is often the loan amount financed.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. A loan can be seen as PV of future repayments.
2. Longer repayment periods lower payment but may increase total interest.
3. PV factor rises with n, holding positive i fixed.
4. PV factor falls as i rises, holding n fixed.

Problem-Solving Routine

- Step 1: Identify known values, unknown value and timing.
- Step 2: Convert rates and time units so periods match.
- Step 3: Choose the formula and substitute carefully.
- Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 11 - PRACTICE

Present Value of an Ordinary Annuity

Common errors, guided practice and mastery check

Common error	Better method	Why?
Using FV annuity formula for a loan.	Use PV annuity formula.	A loan is valued at the start.
Using nominal annual rate as i for monthly payments.	Convert to monthly periodic rate.	Period matching is essential.

Guided Practice

1. State PV ordinary annuity formula.
2. What does the PV factor represent?
3. What happens to PV when i rises for same payments?
4. How do you solve for loan payment R ?
5. Why is a loan principal often an annuity PV?
6. Do end-of-period payments occur at time 0?
7. What happens to payment if n increases for same PV and i ?
8. What is the no-interest PV of n payments R ?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

1. $PV = R[1 - (1+i)^{-n}]/i$.
2. PV of one unit paid each period.
3. PV falls.
4. $R = PV/\text{factor}$.
5. It equals the present value of future repayments.
6. No.
7. Payment falls.
8. Rn .

Independent Practice

- A. Write three new numerical examples based on this lesson.
- B. Change one given rate or time period and predict how the answer changes.
- C. Write one deliberately incorrect calculation, then correct it and explain the error.
- D. Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 12 - CORE FINANCIAL MATHEMATICS

Annuity Due

Concepts, formulas and worked examples

Learning Objectives

- Calculate present and future value of beginning-of-period payments.
- Convert ordinary-annuity values to annuity-due values.
- Recognize due-payment timing.

Concept / Formula	Explanation	Worked Example
FV annuity due	$FV_{\text{due}} = FV_{\text{ordinary}}(1+i)$	Each payment compounds one extra period.
PV annuity due	$PV_{\text{due}} = PV_{\text{ordinary}}(1+i)$	Each payment is one period earlier.
Timing	First payment at time 0, last at time $n-1$.	Beginning-of-period series.
Relative value	For positive i , due value exceeds ordinary value with same R and n .	Earlier cash flows are more valuable.

Key Rules

- Identify timing before selecting formula.
- Multiply ordinary annuity value by $1+i$ for a due annuity at matching valuation dates.
- A payment at time 0 is not discounted.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. Rent paid at the start of each month resembles an annuity due.
2. A lease payment schedule may be due.
3. Due FV exceeds ordinary FV.
4. Due PV exceeds ordinary PV for positive rates.

Problem-Solving Routine

Step 1: Identify known values, unknown value and timing.

Step 2: Convert rates and time units so periods match.

Step 3: Choose the formula and substitute carefully.

Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 12 - PRACTICE

Annuity Due

Common errors, guided practice and mastery check

Common error	Better method	Why?
Using due formula when payments are at month-end.	Use ordinary-annuity formulas.	Timing determines type.
Adding one extra payment accidentally.	Keep n equal to actual number of payments.	The $1+i$ adjustment shifts timing, not count.

Guided Practice

1. What is the first payment time for an annuity due?
2. How convert ordinary FV to due FV?
3. How convert ordinary PV to due PV?
4. Which is larger at positive i : due or ordinary PV?
5. Does due annuity have more payments if n is same?
6. Give one example of due payments.
7. Why is due value larger?
8. What happens when $i=0$?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

1. Time 0.
2. Multiply by $1+i$.
3. Multiply by $1+i$.
4. Due PV.
5. No.
6. For example, rent paid at beginning of month.
7. Each payment occurs one period earlier / compounds longer.
8. Due and ordinary values are equal for same payment count.

Independent Practice

- A. Write three new numerical examples based on this lesson.
- B. Change one given rate or time period and predict how the answer changes.
- C. Write one deliberately incorrect calculation, then correct it and explain the error.
- D. Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 13 - CORE FINANCIAL MATHEMATICS

Perpetuities

Concepts, formulas and worked examples

Learning Objectives

- Define a perpetuity.
- Calculate present value of level perpetuity.
- Understand sensitivity to discount rate.

Concept / Formula	Explanation	Worked Example
Level perpetuity	Infinite stream of equal payments.	Payment R continues forever.
PV formula	$PV = R/i$	Requires $i > 0$ and first payment one period from now.
Required return	$i = R/PV$	Can infer yield from price and payment.
Growth perpetuity	$PV = R1/(i-g)$	Advanced form if payments grow at constant g and $i > g$.

Key Rules

- A standard perpetuity formula assumes first payment at end of first period.
- PV is finite because distant payments are heavily discounted.
- Small changes in i can cause large PV changes.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. $R=100$ annually, $i=5\%$ gives $PV=2000$.
2. At 4%, same R has $PV=2500$.
3. Higher required return lowers value.
4. A growing perpetuity needs $i > g$.

Problem-Solving Routine

- Step 1: Identify known values, unknown value and timing.
- Step 2: Convert rates and time units so periods match.
- Step 3: Choose the formula and substitute carefully.
- Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 13 - PRACTICE

Perpetuities

Common errors, guided practice and mastery check

Common error	Better method	Why?
Using $PV=Ri$.	Use $PV=R/i$.	The payment is divided by the periodic discount rate.
Using $g \geq i$ in growth perpetuity.	Formula requires $i > g$.	Otherwise the infinite sum does not converge under the simple model.

Guided Practice

- Find PV of 120 per year at 6%.
- Find PV of 500 per year at 10%.
- What happens to perpetuity PV when i rises?
- What timing is assumed for first payment?
- Solve i if $R=80$ and $PV=1600$.
- Why can an infinite stream have finite PV?
- State growing perpetuity formula.
- What condition is needed for growing perpetuity?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

- 2000.
- 5000.
- PV falls.
- End of first period.
- 5%.
- Distant cash flows are heavily discounted.
- $PV=R1/(i-g)$.
- $i > g$.

Independent Practice

- Write three new numerical examples based on this lesson.
- Change one given rate or time period and predict how the answer changes.
- Write one deliberately incorrect calculation, then correct it and explain the error.
- Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 14 - CORE FINANCIAL MATHEMATICS

Loans and Amortization

Concepts, formulas and worked examples

Learning Objectives

- Model an installment loan as an annuity.
- Calculate periodic payment.
- Separate interest and principal components.

Concept / Formula	Explanation	Worked Example
Loan payment	$R = PV / a\text{-angle-}n$	For an ordinary level-payment amortizing loan.
Interest portion	Opening balance $\times$ periodic rate	Interest is calculated before principal reduction.
Principal portion	Payment - interest	Amount reducing the loan balance.
Closing balance	Opening balance - principal repaid	Becomes next period opening balance.

Key Rules

- Early payments contain more interest when balance is larger.
- As balance falls, interest portion falls and principal portion rises for fixed payment.
- Final balance should be approximately zero, allowing for rounding.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. A 5-year monthly loan has 60 payments.
2. Monthly rate must match monthly payment frequency.
3. Each payment covers interest plus principal.
4. Total interest = total payments - original principal, ignoring fees.

Problem-Solving Routine

- Step 1: Identify known values, unknown value and timing.
- Step 2: Convert rates and time units so periods match.
- Step 3: Choose the formula and substitute carefully.
- Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 14 - PRACTICE

Loans and Amortization

Common errors, guided practice and mastery check

Common error	Better method	Why?
Subtracting the whole payment before calculating interest.	Calculate interest on opening balance first.	Interest accrues on balance outstanding during the period.
Using annual rate directly for monthly interest.	Convert to monthly periodic rate.	Rate period must match payment interval.

Guided Practice

1. What is the formula concept for level loan payment?
2. How is period interest calculated?
3. How is principal repaid calculated?
4. How is closing balance calculated?
5. What usually happens to interest portion over time?
6. What usually happens to principal portion?
7. How many monthly payments in 3 years?
8. How can total interest be found from payments?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

1. $R = PV/\text{annuity PV factor}$.
2. Opening balance $\times$ periodic rate.
3. Payment - interest.
4. Opening balance - principal repaid.
5. It falls.
6. It rises for a fixed payment.
7. 36.
8. Total of payments minus original principal, ignoring fees.

Independent Practice

- A. Write three new numerical examples based on this lesson.
- B. Change one given rate or time period and predict how the answer changes.
- C. Write one deliberately incorrect calculation, then correct it and explain the error.
- D. Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 15 - CORE FINANCIAL MATHEMATICS

Amortization Tables

Concepts, formulas and worked examples

Learning Objectives

- Construct a loan amortization schedule.
- Track balance, interest and principal.
- Check schedule consistency.

Concept / Formula	Explanation	Worked Example
Opening balance	Balance before each payment.	Starts with original loan amount.
Interest	Opening balance $\times$ i	Periodic finance charge.
Payment	Usually constant R	May be adjusted slightly in final period for rounding.
Principal repaid	$R - \text{interest}$	Reduces balance.
Closing balance	Opening - principal	Next period opening balance.

Key Rules

- Use unrounded internal values where possible.
- The last payment may differ by a few cents because of rounding.
- Sum of principal repayments should equal original loan.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. If opening balance is 1000 and $i=1\%$, interest is 10.
2. If payment is 90, principal repaid is 80.
3. Closing balance becomes 920.
4. Next interest is based on 920, not 1000.

Problem-Solving Routine

- Step 1: Identify known values, unknown value and timing.
- Step 2: Convert rates and time units so periods match.
- Step 3: Choose the formula and substitute carefully.
- Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 15 - PRACTICE

Amortization Tables

Common errors, guided practice and mastery check

Common error	Better method	Why?
Keeping interest constant each period.	Recalculate from current opening balance.	The balance changes after each payment.
Rounding balance too aggressively.	Maintain precision and round displayed money only.	Rounding error can distort final balance.

Guided Practice

1. If balance=2000 and $i=2\%$, find interest.
2. If payment=300, find principal repaid.
3. Find closing balance.
4. What becomes next opening balance?
5. Why does interest usually fall?
6. What should sum of principal repayments equal?
7. Why might final payment be adjusted?
8. What key columns appear in an amortization table?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

1. 40.
2. 260.
3. 1740.
4. 1740.
5. Outstanding balance falls.
6. Original principal.
7. To correct accumulated rounding.
8. Period, opening balance, interest, payment, principal, closing balance.

Independent Practice

- A. Write three new numerical examples based on this lesson.
- B. Change one given rate or time period and predict how the answer changes.
- C. Write one deliberately incorrect calculation, then correct it and explain the error.
- D. Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 16 - CORE FINANCIAL MATHEMATICS

Sinking Funds

Concepts, formulas and worked examples

Learning Objectives

- Explain a sinking fund.
- Calculate periodic deposits for a future target.
- Compare sinking funds with loan amortization.

Concept / Formula	Explanation	Worked Example
Sinking fund	Regular deposits accumulated to meet a future obligation.	Saving to replace equipment.
Deposit formula	$R = FV / s\text{-angle-}n$	Target FV divided by annuity accumulation factor.
Interest earned	$FV - \text{total deposits}$	Growth supplied by investment returns.
Contrast with amortization	Sinking fund balance grows; loan balance falls.	Opposite accumulation direction.

Key Rules

- Match deposit frequency and interest period.
- Ordinary sinking-fund deposits are usually at period end unless stated otherwise.
- Higher return reduces required deposit for same target and horizon.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. Save 10000 over 5 years with monthly deposits.
2. A business can build a replacement reserve.
3. Longer saving horizon lowers required periodic deposit.
4. Interest contributes to target.

Problem-Solving Routine

- Step 1: Identify known values, unknown value and timing.
- Step 2: Convert rates and time units so periods match.
- Step 3: Choose the formula and substitute carefully.
- Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 16 - PRACTICE

Sinking Funds

Common errors, guided practice and mastery check

Common error	Better method	Why?
Dividing target simply by number of payments when interest is positive.	Use annuity FV factor.	Deposits earn interest.
Using PV annuity factor.	Use FV annuity factor for a future target.	The goal is an accumulated amount.

Guided Practice

1. Define a sinking fund.
2. State deposit formula concept.
3. What happens to required R if i rises?
4. What happens to required R if n rises?
5. What does total deposits plus interest equal?
6. Which factor is used: PV or FV annuity?
7. How does a sinking fund balance usually change?
8. Give one business use.

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

1. Regular deposits accumulated for a future obligation.
2. $R = FV / FV \text{ annuity factor}$.
3. R falls.
4. R falls, other things equal.
5. Target future value.
6. FV annuity factor.
7. It grows.
8. Equipment replacement reserve.

Independent Practice

- A. Write three new numerical examples based on this lesson.
- B. Change one given rate or time period and predict how the answer changes.
- C. Write one deliberately incorrect calculation, then correct it and explain the error.
- D. Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 17 - CORE FINANCIAL MATHEMATICS

Depreciation of Assets

Concepts, formulas and worked examples

Learning Objectives

- Use straight-line depreciation.
- Use reducing-balance depreciation.
- Distinguish book value and salvage value.

Concept / Formula	Explanation	Worked Example
Straight-line depreciation	Annual depreciation = $(\text{Cost} - \text{Salvage})/\text{Life}$	Equal depreciation each year.
Book value straight-line	Cost - accumulated depreciation	Falls linearly.
Reducing balance	$BV_n = \text{Cost}(1-r)^n$	Constant percentage applied to current book value.
Salvage value	Estimated value at end of useful life.	Used in straight-line calculation.

Key Rules

- Depreciation is an accounting allocation, not necessarily market-value loss.
- Straight-line gives equal annual expense.
- Reducing balance gives larger depreciation early and smaller later.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. Cost 10000, salvage 2000, life 4 years gives 2000/year straight-line.
2. At 20% reducing balance, year-end factor is 0.8.
3. Book value cannot normally be interpreted as guaranteed resale value.
4. Accumulated depreciation is sum of annual charges.

Problem-Solving Routine

- Step 1: Identify known values, unknown value and timing.
- Step 2: Convert rates and time units so periods match.
- Step 3: Choose the formula and substitute carefully.
- Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 17 - PRACTICE

Depreciation of Assets

Common errors, guided practice and mastery check

Common error	Better method	Why?
Applying straight-line rate to current book value.	Use fixed annual amount for straight-line.	Only reducing balance applies a rate to current BV.
Forgetting salvage value in straight-line.	Subtract salvage before dividing by life.	Only depreciable amount is allocated.

Guided Practice

- Find straight-line depreciation for cost 15000, salvage 3000, life 6 years.
- Find BV after 2 years under that method.
- Find BV after 3 years for cost 10000 at 15% reducing balance.
- Which method gives equal annual depreciation?
- Which method gives higher early depreciation?
- Define book value.
- Define salvage value.
- Is book value always market value?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

- 2000/year.
- 11000.
- 6141.25.
- Straight-line.
- Reducing balance.
- Cost less accumulated depreciation under the accounting method.
- Estimated end-of-life value.
- No.

Independent Practice

- Write three new numerical examples based on this lesson.
- Change one given rate or time period and predict how the answer changes.
- Write one deliberately incorrect calculation, then correct it and explain the error.
- Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 18 - CORE FINANCIAL MATHEMATICS

Inflation and Purchasing Power

Concepts, formulas and worked examples

Learning Objectives

- Calculate price changes under inflation.
- Distinguish nominal and real values.
- Estimate loss of purchasing power.

Concept / Formula	Explanation	Worked Example
Inflated price	Future price = current price $(1+f)^n$	f is inflation rate.
Purchasing power factor	$1/(1+f)^n$	Shows how much future money is worth in today's buying power.
Approx real return	Nominal return - inflation	Useful rough estimate for small rates.
Exact real return	$(1+\text{nominal})/(1+\text{inflation})-1$	More accurate relation.

Key Rules

- Inflation reduces purchasing power of money.
- Nominal growth does not guarantee real growth.
- Use compound inflation over multiple years.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. 100 at 5% inflation for 3 years becomes about 115.76.
2. A 7% investment return with 4% inflation has exact real return about 2.88%.
3. Real values compare buying power.
4. Inflation assumptions matter in long-term planning.

Problem-Solving Routine

Step 1: Identify known values, unknown value and timing.

Step 2: Convert rates and time units so periods match.

Step 3: Choose the formula and substitute carefully.

Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 18 - PRACTICE

Inflation and Purchasing Power

Common errors, guided practice and mastery check

Common error	Better method	Why?
Subtracting total inflation as $n \times f$ without context.	Use compound factor $(1+f)^n$ for repeated annual inflation.	Inflation compounds.
Calling nominal return the real return.	Adjust for inflation.	Purchasing power matters.

Guided Practice

- Find future price of 500 after 2 years at 6% inflation.
- Find exact real return if nominal=8%, inflation=3%.
- What happens to purchasing power under positive inflation?
- What is rough real return for 10% nominal and 4% inflation?
- State exact real return formula.
- Why can cash lose real value even if amount is unchanged?
- Does inflation compound over years?
- What does a real return measure?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

- 561.80.
- 4.854% approximately.
- It falls.
- About 6%.
- $(1+\text{nominal})/(1+\text{inflation})-1$.
- Prices rise, so the same cash buys less.
- Yes.
- Growth in purchasing power.

Independent Practice

- Write three new numerical examples based on this lesson.
- Change one given rate or time period and predict how the answer changes.
- Write one deliberately incorrect calculation, then correct it and explain the error.
- Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 19 - CORE FINANCIAL MATHEMATICS

Investment Returns

Concepts, formulas and worked examples

Learning Objectives

- Calculate holding period return.
- Include income and capital gain.
- Annualize simple multi-period returns carefully.

Concept / Formula	Explanation	Worked Example
Holding period return	$HPR = (Ending\ value - Beginning\ value + Income) / Beginning\ value$	Total return over holding period.
Capital gain	Ending value - Beginning value	Price appreciation component.
Income return	Income / Beginning value	Dividend or interest component.
Geometric annual return	$(Ending/Beginning)^{(1/n)} - 1$ when no cash flows	Compound annual growth rate.

Key Rules

- Include both income and price change in total return.
- Use geometric annualization for compound growth.
- Do not average percentage returns arithmetically when compounding across periods unless asked for arithmetic mean.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. Buy at 100, receive 4 dividend, sell at 108: $HPR=12\%$.
2. CAGR from 100 to 121 in 2 years is 10%.
3. Negative capital gain can be offset by income.
4. Return percentages need a clear time period.

Problem-Solving Routine

- Step 1: Identify known values, unknown value and timing.
- Step 2: Convert rates and time units so periods match.
- Step 3: Choose the formula and substitute carefully.
- Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 19 - PRACTICE

Investment Returns

Common errors, guided practice and mastery check

Common error	Better method	Why?
Ignoring dividends or interest income.	Include income in total return.	Total return has both price and income components.
Using $(121-100)/100/2$ as exact annual compound return.	Use geometric annual rate.	Compounding is multiplicative.

Guided Practice

- Buy at 50, dividend 2, sell at 55. Find HPR.
- Find capital gain return only.
- Find income return only.
- Find CAGR from 200 to 242 over 2 years.
- What components form total return?
- Can HPR be negative?
- Why specify holding period?
- Which mean fits compound growth across years?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

- 14%.
- 10%.
- 4%.
- 10%.
- Capital gain plus income.
- Yes.
- Return depends on time interval.
- Geometric mean / CAGR.

Independent Practice

- Write three new numerical examples based on this lesson.
- Change one given rate or time period and predict how the answer changes.
- Write one deliberately incorrect calculation, then correct it and explain the error.
- Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 20 - CORE FINANCIAL MATHEMATICS

Net Present Value (NPV)

Concepts, formulas and worked examples

Learning Objectives

- Calculate NPV of a simple project.
- Interpret discount rate and cash flows.
- Use NPV as a value measure.

Concept / Formula	Explanation	Worked Example
NPV	$NPV = \sum CF_t / (1+r)^t$ including CF_0	Initial investment is often negative at $t=0$.
Discount rate	Required return / opportunity cost under the model.	Used to discount future cash flows.
Positive NPV	PV of benefits exceeds PV of costs.	Project adds value under assumptions.
Negative NPV	PV of benefits is below PV of costs.	Project falls short of required return.

Key Rules

- Discount each future cash flow separately.
- Time-0 cash flow is not discounted.
- NPV conclusions depend on forecast cash flows and discount rate.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. $CF_0 = -1000$, $CF_1 = 1100$, $r = 10\%$ gives $NPV = 0$.
2. Higher discount rates reduce PV of positive future cash flows.
3. NPV is measured in money, not percentage.
4. Sensitivity analysis can test uncertain assumptions.

Problem-Solving Routine

- Step 1: Identify known values, unknown value and timing.
- Step 2: Convert rates and time units so periods match.
- Step 3: Choose the formula and substitute carefully.
- Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 20 - PRACTICE

Net Present Value (NPV)

Common errors, guided practice and mastery check

Common error	Better method	Why?
Adding future cash flows without discounting.	Discount each to time 0.	Timing affects value.
Treating NPV as an interest rate.	NPV is a monetary value.	IRR is the rate concept.

Guided Practice

1. State NPV formula concept.
2. Is CF0 discounted?
3. What does positive NPV mean under the model?
4. What happens to NPV when r rises for positive future cash flows?
5. Is NPV a percentage?
6. Why perform sensitivity analysis?
7. Find NPV for -1000 now and +1210 in 2 years at 10%.
8. What sign is an initial investment usually?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

1. Sum of discounted cash flows including time 0.
2. No.
3. PV of benefits exceeds costs at the required return.
4. It generally falls.
5. No, it is money.
6. To see how results change with uncertain inputs.
7. 0.
8. Negative.

Independent Practice

- A. Write three new numerical examples based on this lesson.
- B. Change one given rate or time period and predict how the answer changes.
- C. Write one deliberately incorrect calculation, then correct it and explain the error.
- D. Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 21 - PRACTICE

Internal Rate of Return (IRR)

Common errors, guided practice and mastery check

Common error	Better method	Why?
Calling the discount rate used in NPV automatically the IRR.	IRR is specifically the rate that sets NPV to zero.	It is solved from the cash flows.
Assuming every project has one unique IRR.	Check cash-flow pattern.	Multiple sign changes can create multiple solutions.

Guided Practice

1. Define IRR.
2. Find IRR for -1000 now and +1200 in 1 year.
3. What is NPV at the IRR?
4. Is IRR a money value or rate?
5. Can a project have multiple IRRs?
6. Why might IRR and NPV rankings differ?
7. What numerical issue arises with complex cash flows?
8. What measure complements IRR well?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

1. Rate making NPV zero.
2. 20%.
3. Zero.
4. A rate.
5. Yes, for some nonconventional cash flows.
6. Scale/timing differences can matter.
7. IRR may need iterative numerical solving.
8. NPV.

Independent Practice

- A. Write three new numerical examples based on this lesson.
- B. Change one given rate or time period and predict how the answer changes.
- C. Write one deliberately incorrect calculation, then correct it and explain the error.
- D. Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 22 - CORE FINANCIAL MATHEMATICS

Bond Pricing

Concepts, formulas and worked examples

Learning Objectives

- Value a bond as coupons plus redemption value.
- Understand coupon rate, face value and yield.
- Explain inverse price-yield relation.

Concept / Formula	Explanation	Worked Example
Coupon payment	$C = \text{coupon rate} \times \text{face value per coupon period}$	A 6% annual coupon on 1000 pays 60 per year.
Bond price	PV of coupons + PV of face value	$P = C a_{\overline{n} i} + F/(1+i)^n$ for level coupons.
Face value	Amount repaid at maturity.	Often 100 or 1000.
Yield	Discount rate consistent with market price and cash flows.	Higher yield means lower price, other things equal.

Key Rules

- Match coupon and yield periods.
- Bond price equals face value when coupon rate equals yield under matching conventions.
- Price and yield move inversely.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. A 5% coupon bond on face 1000 pays 50 annually.
2. Discount coupons as an annuity and redemption as a lump sum.
3. If yield exceeds coupon rate, price is usually below face value.
4. Longer maturity generally increases interest-rate sensitivity.

Problem-Solving Routine

- Step 1: Identify known values, unknown value and timing.
- Step 2: Convert rates and time units so periods match.
- Step 3: Choose the formula and substitute carefully.
- Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 22 - PRACTICE

Bond Pricing

Common errors, guided practice and mastery check

Common error	Better method	Why?
Discounting only the face value.	Discount both coupons and redemption.	Both are future cash flows.
Comparing annual coupon rate to semiannual yield without conversion.	Use consistent period rates.	Frequency matters.

Guided Practice

- Find annual coupon on face 2000 at 4%.
- What are the two components of bond price?
- What happens to price when yield rises?
- When is price approximately at par?
- If yield > coupon rate, is price usually above or below par?
- Why match periods?
- What is face value?
- What happens to interest-rate sensitivity with longer maturity, generally?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

- 80.
- PV of coupons and PV of redemption value.
- Price falls.
- When coupon rate equals yield under matching conventions.
- Below par.
- Coupon and discount periods must be consistent.
- Amount repaid at maturity.
- It generally increases.

Independent Practice

- Write three new numerical examples based on this lesson.
- Change one given rate or time period and predict how the answer changes.
- Write one deliberately incorrect calculation, then correct it and explain the error.
- Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 23 - CORE FINANCIAL MATHEMATICS

Shares, Dividends and Basic Valuation

Concepts, formulas and worked examples

Learning Objectives

- Calculate dividend yield.
- Use simple dividend-growth valuation.
- Distinguish market price and book/accounting measures conceptually.

Concept / Formula	Explanation	Worked Example
Dividend yield	Annual dividend per share / market price per share	Dividend 2, price 40 gives 5%.
Capital gain yield	$(P_1 - P_0)/P_0$ over period	Price from 40 to 44 gives 10%.
Total shareholder return	Dividend yield + capital gain yield for same period, approximately/exactly with common base	Dividend 2 and gain 4 on initial 40 gives 15%.
Constant-growth model	$P_0 = D_1/(k-g)$, $k > g$	Simple valuation model for perpetually growing dividends.

Key Rules

- Dividend yield uses current/initial market price as specified.
- Share valuation models rely on assumptions and are sensitive to growth and required return.
- Total return should include both dividends and price change.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. A share bought at 20, dividend 1, sold at 22 gives 15% total HPR.
2. If expected $D_1=3$, $k=10\%$, $g=4\%$, model price=50.
3. Higher k lowers model value.
4. Higher sustainable g raises model value if $k > g$.

Problem-Solving Routine

Step 1: Identify known values, unknown value and timing.

Step 2: Convert rates and time units so periods match.

Step 3: Choose the formula and substitute carefully.

Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 23 - PRACTICE

Shares, Dividends and Basic Valuation

Common errors, guided practice and mastery check

Common error	Better method	Why?
Ignoring dividends in stock return.	Include cash dividends in total return.	Investors receive both income and capital gain/loss.
Using $g > k$ in constant-growth formula.	Require $k > g$.	Otherwise the simple perpetuity model does not converge.

Guided Practice

- Find dividend yield if dividend=1.5 and price=30.
- Find capital gain yield if price rises 30 to 33.
- Find total HPR including dividend 1.5.
- State constant-growth formula.
- What condition is required?
- What happens to value if k rises?
- What happens to value if g rises, other things equal and $k > g$?
- Does dividend yield equal total return in general?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

- 5%.
- 10%.
- 15%.
- $P_0 = D_1 / (k - g)$.
- $k > g$.
- Value falls.
- Value rises.
- No.

Independent Practice

- Write three new numerical examples based on this lesson.
- Change one given rate or time period and predict how the answer changes.
- Write one deliberately incorrect calculation, then correct it and explain the error.
- Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

Concepts, formulas and worked examples

- Interpret exchange-rate quotations.
- Convert currencies correctly.
- Calculate percentage currency changes.

Key Rules

- Always write units beside exchange rates.
- Use dimensional reasoning to decide multiply or divide.
- Real transactions may include spreads and fees.

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

1. 500 USD at 4500 MGA/USD gives 2,250,000 MGA before fees.
2. To convert 900,000 MGA to USD at 4500 MGA/USD, divide to get 200 USD.
3. A stronger domestic currency means fewer domestic units per foreign unit under this quote.
4. Spreads create transaction costs.

Step 1: Identify known values, unknown value and timing.

Step 2: Convert rates and time units so periods match.

Step 3: Choose the formula and substitute carefully.

Step 4: Interpret the numerical result in words and check reasonableness.

Create one similar problem, solve it, and underline the formula used.

LESSON 24 - PRACTICE

Foreign Exchange Mathematics

Common errors, guided practice and mastery check

Common error	Better method	Why?
Multiplying when units do not cancel correctly.	Write units and choose operation so unwanted units cancel.	Dimensional analysis prevents many errors.
Ignoring bid/ask direction.	Use the rate applicable to buying or selling.	Dealer quotes can differ.

Guided Practice

- Convert 250 USD at 4600 MGA/USD to MGA.
- Convert 920000 MGA at 4600 MGA/USD to USD.
- What is inverse of 5000 MGA/USD in USD/MGA?
- Why are units important?
- What is a spread?
- Do fees affect actual converted amount?
- If quote rises from 4500 to 4725, find percentage rise.
- Under MGA/USD quote, what does a higher quote mean for MGA relative to USD?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

- 1,150,000 MGA.
- 200 USD.
- 0.0002 USD/MGA.
- They show whether to multiply or divide.
- Difference between buying and selling rates.
- Yes.
- 5%.
- MGA has weakened relative to USD.

Independent Practice

- Write three new numerical examples based on this lesson.
- Change one given rate or time period and predict how the answer changes.
- Write one deliberately incorrect calculation, then correct it and explain the error.
- Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 25 - CORE FINANCIAL MATHEMATICS

Payroll, Wages and Deductions

Concepts, formulas and worked examples

Learning Objectives

- Calculate gross pay.
- Apply overtime and percentage deductions.
- Find net pay.

Concept / Formula	Explanation	Worked Example
Gross pay	Regular pay + overtime + allowances	Before deductions.
Hourly pay	Hours x hourly rate	Base wage calculation.
Overtime	Overtime hours x overtime rate	Rate may be 1.5x normal rate.
Net pay	Gross pay - deductions	Take-home pay before/after any specified adjustments.

Key Rules

- Apply deduction percentages to the correct base.
- Distinguish gross, taxable and net pay if a problem provides different definitions.
- Use overtime multiplier only for overtime hours.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. 40 hours at 10 plus 5 overtime hours at 15 gives gross 475.
2. A 6% deduction on gross 475 is 28.50.
3. Net after that deduction is 446.50.
4. Allowances may be fixed or percentage-based depending on rules.

Problem-Solving Routine

- Step 1: Identify known values, unknown value and timing.
- Step 2: Convert rates and time units so periods match.
- Step 3: Choose the formula and substitute carefully.
- Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 25 - PRACTICE

Payroll, Wages and Deductions

Common errors, guided practice and mastery check

Common error	Better method	Why?
Applying overtime rate to all hours.	Apply it only to overtime hours.	Regular hours use regular rate.
Subtracting 6 instead of 6%.	Convert 6% to 0.06 times the stated base.	Percentage deductions are proportional.

Guided Practice

- Find gross pay: 40 h at 12 plus 4 h overtime at 18.
- Find 5% deduction on gross 552.
- Find net after that deduction.
- What is gross pay?
- What is net pay?
- What does 1.5x overtime mean?
- Why identify deduction base?
- Can allowances increase gross pay?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

- 552.
- 27.60.
- 524.40.
- Pay before deductions.
- Pay after specified deductions.
- One and a half times normal hourly rate.
- Different deductions may apply to different bases.
- Yes.

Independent Practice

- Write three new numerical examples based on this lesson.
- Change one given rate or time period and predict how the answer changes.
- Write one deliberately incorrect calculation, then correct it and explain the error.
- Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 26 - CORE FINANCIAL MATHEMATICS

Markup, Markdown and Profit

Concepts, formulas and worked examples

Learning Objectives

- Calculate markup on cost and on selling price.
- Calculate markdown and sale price.
- Distinguish profit margin from markup rate.

Concept / Formula	Explanation	Worked Example
Profit	Selling price - Cost	Basic trading profit before other expenses.
Markup on cost	Profit / Cost	Cost 80, sell 100 -> 25% markup on cost.
Profit margin	Profit / Selling price	Same example -> 20% margin.
Markdown	Original price x markdown rate	20% off 150 means discount 30 and sale price 120.

Key Rules

- Markup percentage depends on its base.
- Margin and markup are not equal except in trivial cases.
- Successive discounts multiply, not simply add in general.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. Cost 60 with 40% markup on cost gives price 84.
2. Price 200 with 15% markdown gives 170.
3. Profit 30 on sales 120 gives margin 25%.
4. Two discounts 10% then 20% give factor $0.9 \times 0.8 = 0.72$, total 28% off.

Problem-Solving Routine

- Step 1: Identify known values, unknown value and timing.
- Step 2: Convert rates and time units so periods match.
- Step 3: Choose the formula and substitute carefully.
- Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 26 - PRACTICE

Markup, Markdown and Profit

Common errors, guided practice and mastery check

Common error	Better method	Why?
Using selling price as denominator for markup on cost.	Use cost in denominator.	The phrase "on cost" fixes the base.
Adding successive discounts $10\%+20\%=30\%$.	Use multipliers.	Second discount applies to already-reduced price.

Guided Practice

- Cost=80, sell=104. Find profit.
- Find markup on cost.
- Find profit margin.
- Original 500, markdown 12%. Find sale price.
- Apply successive discounts 10% then 20% to 100.
- What is difference between markup and margin?
- If markup on cost is 50%, what multiplier gives selling price?
- Can margin exceed 100% with positive cost and price?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

- 24.
- 30%.
- 23.0769% approximately.
- 440.
- 72.
- Different denominator bases.
- 1.5.
- No under ordinary positive-cost sales.

Independent Practice

- Write three new numerical examples based on this lesson.
- Change one given rate or time period and predict how the answer changes.
- Write one deliberately incorrect calculation, then correct it and explain the error.
- Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 27 - CORE FINANCIAL MATHEMATICS

Break-Even Analysis

Concepts, formulas and worked examples

Learning Objectives

- Separate fixed and variable costs.
- Calculate contribution and break-even quantity.
- Use break-even for planning.

Concept / Formula	Explanation	Worked Example
Fixed cost F	Cost not changing directly with output in the modeled range.	Rent.
Variable cost v	Cost per unit that changes with output.	Materials per item.
Contribution per unit	$p-v$	Amount each unit contributes toward fixed cost and profit.
Break-even quantity	$Q_{BE} = F/(p-v)$	Units where profit is zero.

Key Rules

- Break-even requires $p > v$.
- Profit = $(p-v)Q - F$ in the linear model.
- Actual businesses may have nonlinear costs and capacity limits.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. $F=10000$, $p=50$, $v=30$ gives contribution 20 and break-even 500 units.
2. Above break-even, each extra unit adds contribution to profit in the simple model.
3. At break-even, total revenue equals total cost.
4. Margin of safety compares actual sales with break-even sales.

Problem-Solving Routine

- Step 1: Identify known values, unknown value and timing.
- Step 2: Convert rates and time units so periods match.
- Step 3: Choose the formula and substitute carefully.
- Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 27 - PRACTICE

Break-Even Analysis

Common errors, guided practice and mastery check

Common error	Better method	Why?
Dividing fixed cost by price.	Divide by contribution $p-v$.	Variable cost must be covered first.
Calling break-even the maximum profit point.	Break-even is zero-profit point.	Profit becomes positive only beyond it, in the simple model.

Guided Practice

- Find contribution if $p=75$, $v=45$.
- Find break-even units if $F=12000$.
- Find profit at $Q=500$ using same data.
- What condition is needed for break-even formula to make sense?
- At break-even, what is profit?
- Define fixed cost.
- Define variable cost.
- What is margin of safety conceptually?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

- 30.
- 400 units.
- 3000.
- $p > v$.
- Zero.
- Cost not changing directly with output in modeled range.
- Cost changing with activity/output.
- Amount sales exceed break-even level.

Independent Practice

- Write three new numerical examples based on this lesson.
- Change one given rate or time period and predict how the answer changes.
- Write one deliberately incorrect calculation, then correct it and explain the error.
- Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 28 - CORE FINANCIAL MATHEMATICS

Taxes, Discounts and Consumer Finance

Concepts, formulas and worked examples

Learning Objectives

- Calculate sales tax or VAT-style additions.
- Reverse a tax-inclusive price.
- Handle discounts before tax when specified.

Concept / Formula	Explanation	Worked Example
Tax added	$\text{Final} = \text{pre-tax} \times (1+t)$	100 plus 20% tax = 120.
Tax amount	$\text{Pre-tax} \times t$	At 20%, tax on 100 is 20.
Reverse tax	$\text{Pre-tax} = \text{tax-inclusive} / (1+t)$	120 inclusive at 20% $\rightarrow$ 100 pre-tax.
Discount then tax	$\text{Price} \times (1-d)(1+t)$	Order follows transaction rules.

Key Rules

- Read whether tax is included or added.
- To remove included tax, divide by $1+t$, not subtract $t\%$ of the inclusive price.
- Apply discounts and taxes in the stated order.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. Inclusive 240 at 20% tax has pre-tax price 200.
2. 10% discount then 20% tax on 100 gives 108.
3. A 20% tax-inclusive price contains tax equal to $20/120$ of the final price, not $20/100$.
4. Consumer-finance calculations depend on contractual definitions.

Problem-Solving Routine

- Step 1: Identify known values, unknown value and timing.
- Step 2: Convert rates and time units so periods match.
- Step 3: Choose the formula and substitute carefully.
- Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 28 - PRACTICE

Taxes, Discounts and Consumer Finance

Common errors, guided practice and mastery check

Common error	Better method	Why?
Subtracting 20% from a tax-inclusive amount to remove 20% tax.	Divide by 1.20.	The tax was 20% of pre-tax price.
Combining discount and tax by netting percentages blindly.	Use sequential multipliers.	They apply to different bases.

Guided Practice

- Find final price of 250 plus 20% tax.
- Find pre-tax price if final is 360 inclusive at 20%.
- Find tax amount in that 360.
- Find final price after 10% discount then 20% tax on 500.
- Why divide to remove included tax?
- What multiplier adds 15% tax?
- What multiplier applies 25% discount?
- Does order of discount and tax matter under percentage-only operations on same base?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

- 300.
- 300.
- 60.
- 540.
- Because tax is a percentage of pre-tax price.
- 1.15.
- 0.75.
- Pure percentage multiplication commutes, though legal bases/rules can differ.

Independent Practice

- Write three new numerical examples based on this lesson.
- Change one given rate or time period and predict how the answer changes.
- Write one deliberately incorrect calculation, then correct it and explain the error.
- Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 29 - CORE FINANCIAL MATHEMATICS

Probability and Financial Risk

Concepts, formulas and worked examples

Learning Objectives

- Calculate expected value in simple risk models.
- Understand variance concept qualitatively.
- Compare risky choices using probabilities and payoffs.

Concept / Formula	Explanation	Worked Example
Expected value	$E(X) = \sum p_i x_i$	Probability-weighted average payoff.
Probability rule	Probabilities sum to 1 for complete mutually exclusive outcomes.	$0.4 + 0.6 = 1$.
Expected return	$\sum p_i r_i$	Weighted average of possible returns.
Risk	Uncertainty of outcomes; dispersion matters as well as mean.	Two investments can share same expected return but differ in variability.

Key Rules

- Expected value is a long-run average, not a guaranteed outcome.
- Probabilities must be consistent and sum to 1 over complete outcomes.
- Higher expected return does not automatically mean better for every decision-maker.

Study Tip

Write the timeline or identify the percentage base before substituting numbers. Keep rates and time periods consistent, then check whether the answer is financially reasonable.

Worked Model Statements

1. 50% chance of +20 and 50% chance of -10 gives $EV=5$.
2. Expected return of $0.3(10\%) + 0.7(4\%) = 5.8\%$.
3. Variance measures spread around the mean.
4. Diversification can reduce some forms of risk but not all.

Problem-Solving Routine

Step 1: Identify known values, unknown value and timing.

Step 2: Convert rates and time units so periods match.

Step 3: Choose the formula and substitute carefully.

Step 4: Interpret the numerical result in words and check reasonableness.

Your Example

Create one similar problem, solve it, and underline the formula used.

LESSON 29 - PRACTICE

Probability and Financial Risk

Common errors, guided practice and mastery check

Common error	Better method	Why?
Choosing the most likely outcome as the expected value.	Weight every outcome by probability.	Expected value uses all outcomes.
Assuming positive EV guarantees profit.	EV is an average across repeated comparable trials.	Single outcomes remain uncertain.

Guided Practice

- Find EV of 0.6 chance 100 and 0.4 chance -20.
- Do probabilities in a complete outcome set sum to what?
- Find expected return: 0.5 at 8%, 0.5 at 2%.
- Is EV guaranteed?
- What does variance describe?
- Can two investments have same mean but different risk?
- What does diversification aim to reduce?
- Why consider both return and risk?

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

- 52.
- 1.
- 5%.
- No.
- Dispersion of outcomes around the mean.
- Yes.
- Some idiosyncratic/unsystematic risk.
- Because outcomes with same average can have different uncertainty.

Independent Practice

- Write three new numerical examples based on this lesson.
- Change one given rate or time period and predict how the answer changes.
- Write one deliberately incorrect calculation, then correct it and explain the error.
- Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

LESSON 30 - PRACTICE

Financial Planning and Integrated Review

Common errors, guided practice and mastery check

Common error	Better method	Why?
Choosing a formula before drawing the timeline.	First identify timing and rate period.	Most errors come from mismatched timing or rates.
Ignoring inflation in a long-term nominal target.	Estimate future cost or use real values.	Purchasing power changes over time.

Guided Practice

1. What should you identify first in a time-value problem?
2. Why compare effective rates?
3. Why use a timeline?
4. Why consider inflation for long goals?
5. What is sensitivity analysis?
6. What is one reason a model may be mathematically correct but practically poor?
7. What is total loan interest approximately equal to for a level-payment loan without fees?
8. Name three major areas combined in financial planning.

Mastery Challenge

- Solve two original problems using the main formula from this lesson.
- Show every rate conversion and time conversion.
- Round final money answers sensibly and state units.

Quick Answers

1. Cash-flow timing, rate convention and objective.
2. To put products on a comparable basis.
3. To locate each cash flow and period.
4. Future prices and purchasing power change.
5. Testing how results change when assumptions change.
6. Unrealistic assumptions or omitted costs/risks.
7. Total payments minus original principal.
8. Saving/investing, borrowing, budgeting/risk/goal planning.

Independent Practice

- A. Write three new numerical examples based on this lesson.
- B. Change one given rate or time period and predict how the answer changes.
- C. Write one deliberately incorrect calculation, then correct it and explain the error.
- D. Apply the concept to a 40-60 word real-life financial context.

Exam Strategy

- Underline what the question asks for: rate, time, present value, future value, payment, profit or percentage.
- Write formulas before substitution when marks are awarded for method.
- Check units, decimal rates and reasonableness before finalizing.

Notes & Corrections

Record one mistake, the correction and the rule that prevents it.

REVISION 01

Review Test 1 - Percentages & Interest

Mixed practice - complete without looking back at the lessons.

1. Percentages and Financial Ratios	Find 18% of 350.
2. Simple Interest	Find simple interest on 2400 at 7% for 3 years.
3. Compound Interest	Find amount on 1500 at 8% for 4 years.
4. Compounding Frequency	Find monthly rate for 9.6% nominal compounded monthly.
5. Percentages and Financial Ratios	Increase 640 by 12%.
6. Simple Interest	Find maturity value of 5000 at 4.5% for 2 years.
7. Compound Interest	Find compound interest on 4000 at 5% for 3 years.
8. Compounding Frequency	Find quarterly rate for 8% nominal quarterly.
9. Percentages and Financial Ratios	Reduce 900 by 25%.
10. Simple Interest	Find interest on 1800 at 10% for 9 months.
11. Compound Interest	Find amount on 12000 at 3% for 10 years.
12. Compounding Frequency	How many monthly periods in 4 years?

Written Task

Explain one financial decision using at least four concepts from this test. Include at least one formula and one interpretation sentence.

Self-Assessment

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Working Space

REVISION 02

Review Test 2 - Rates, PV & Discounting

Mixed practice - complete without looking back at the lessons.

1. Effective and Nominal Interest Rates	Find EAR for 12% nominal monthly.
2. Present Value and Discounting	Find PV of 1331 due in 3 years at 10%.
3. Simple Discount and Bank Discount	Find D on F=8000 at 7% for 9 months.
4. Equivalent Rates and Equation of Value	What is a focal date?
5. Effective and Nominal Interest Rates	Find EAR for 8% nominal quarterly.
6. Present Value and Discounting	Find PV of 5000 due in 5 years at 6%.
7. Simple Discount and Bank Discount	Find proceeds for the same note.
8. Equivalent Rates and Equation of Value	Why move cash flows to one date?
9. Effective and Nominal Interest Rates	Find one-year growth factor for EAR 7.5%.
10. Present Value and Discounting	What happens to PV if r rises?
11. Simple Discount and Bank Discount	Define face value.
12. Equivalent Rates and Equation of Value	How do you move cash forward one year?

Written Task

Explain one financial decision using at least four concepts from this test. Include at least one formula and one interpretation sentence.

Self-Assessment

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Working Space

REVISION 03

Review Test 3 - Annuities & Perpetuities

Mixed practice - complete without looking back at the lessons.

1. Annuities: Basic Concepts	Define an annuity.
2. Future Value of an Ordinary Annuity	State FV ordinary annuity formula.
3. Present Value of an Ordinary Annuity	State PV ordinary annuity formula.
4. Annuity Due	What is the first payment time for an annuity due?
5. Perpetuities	Find PV of 120 per year at 6%.
6. Annuities: Basic Concepts	What is an ordinary annuity?
7. Future Value of an Ordinary Annuity	Find factor for $i=0$ if $n=20$.
8. Present Value of an Ordinary Annuity	What does the PV factor represent?
9. Annuity Due	How convert ordinary FV to due FV?
10. Perpetuities	Find PV of 500 per year at 10%.
11. Annuities: Basic Concepts	What is an annuity due?
12. Future Value of an Ordinary Annuity	What does R represent?

Written Task

Explain one financial decision using at least four concepts from this test. Include at least one formula and one interpretation sentence.

Self-Assessment

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Working Space

REVISION 04

Review Test 4 - Loans & Sinking Funds

Mixed practice - complete without looking back at the lessons.

1. Loans and Amortization	What is the formula concept for level loan payment?
2. Amortization Tables	If balance=2000 and $i=2\%$, find interest.
3. Sinking Funds	Define a sinking fund.
4. Loans and Amortization	How is period interest calculated?
5. Amortization Tables	If payment=300, find principal repaid.
6. Sinking Funds	State deposit formula concept.
7. Loans and Amortization	How is principal repaid calculated?
8. Amortization Tables	Find closing balance.
9. Sinking Funds	What happens to required R if i rises?
10. Loans and Amortization	How is closing balance calculated?
11. Amortization Tables	What becomes next opening balance?
12. Sinking Funds	What happens to required R if n rises?

Written Task

Explain one financial decision using at least four concepts from this test. Include at least one formula and one interpretation sentence.

Self-Assessment

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Working Space

REVISION 05

Review Test 5 - Depreciation, Inflation & Returns

Mixed practice - complete without looking back at the lessons.

1. Depreciation of Assets	Find straight-line depreciation for cost 15000, salvage 3000, life 6 years.
2. Inflation and Purchasing Power	Find future price of 500 after 2 years at 6% inflation.
3. Investment Returns	Buy at 50, dividend 2, sell at 55. Find HPR.
4. Depreciation of Assets	Find BV after 2 years under that method.
5. Inflation and Purchasing Power	Find exact real return if nominal=8%, inflation=3%.
6. Investment Returns	Find capital gain return only.
7. Depreciation of Assets	Find BV after 3 years for cost 10000 at 15% reducing balance.
8. Inflation and Purchasing Power	What happens to purchasing power under positive inflation?
9. Investment Returns	Find income return only.
10. Depreciation of Assets	Which method gives equal annual depreciation?
11. Inflation and Purchasing Power	What is rough real return for 10% nominal and 4% inflation?
12. Investment Returns	Find CAGR from 200 to 242 over 2 years.

Written Task

Explain one financial decision using at least four concepts from this test. Include at least one formula and one interpretation sentence.

Self-Assessment

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Working Space

REVISION 06

Review Test 6 - NPV, IRR & Bonds

Mixed practice - complete without looking back at the lessons.

1. Net Present Value (NPV)	State NPV formula concept.
2. Internal Rate of Return (IRR)	Define IRR.
3. Bond Pricing	Find annual coupon on face 2000 at 4%.
4. Shares, Dividends and Basic Valuation	Find dividend yield if dividend=1.5 and price=30.
5. Net Present Value (NPV)	Is CF0 discounted?
6. Internal Rate of Return (IRR)	Find IRR for -1000 now and +1200 in 1 year.
7. Bond Pricing	What are the two components of bond price?
8. Shares, Dividends and Basic Valuation	Find capital gain yield if price rises 30 to 33.
9. Net Present Value (NPV)	What does positive NPV mean under the model?
10. Internal Rate of Return (IRR)	What is NPV at the IRR?
11. Bond Pricing	What happens to price when yield rises?
12. Shares, Dividends and Basic Valuation	Find total HPR including dividend 1.5.

Written Task

Explain one financial decision using at least four concepts from this test. Include at least one formula and one interpretation sentence.

Self-Assessment

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Working Space

REVISION 07

Review Test 7 - Shares, FX & Payroll

Mixed practice - complete without looking back at the lessons.

1. Shares, Dividends and Basic Valuation	Find dividend yield if dividend=1.5 and price=30.
2. Foreign Exchange Mathematics	Convert 250 USD at 4600 MGA/USD to MGA.
3. Payroll, Wages and Deductions	Find gross pay: 40 h at 12 plus 4 h overtime at 18.
4. Shares, Dividends and Basic Valuation	Find capital gain yield if price rises 30 to 33.
5. Foreign Exchange Mathematics	Convert 920000 MGA at 4600 MGA/USD to USD.
6. Payroll, Wages and Deductions	Find 5% deduction on gross 552.
7. Shares, Dividends and Basic Valuation	Find total HPR including dividend 1.5.
8. Foreign Exchange Mathematics	What is inverse of 5000 MGA/USD in USD/MGA?
9. Payroll, Wages and Deductions	Find net after that deduction.
10. Shares, Dividends and Basic Valuation	State constant-growth formula.
11. Foreign Exchange Mathematics	Why are units important?
12. Payroll, Wages and Deductions	What is gross pay?

Written Task

Explain one financial decision using at least four concepts from this test. Include at least one formula and one interpretation sentence.

Self-Assessment

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Working Space

REVISION 08

Review Test 8 - Markup, Break-Even, Tax & Risk

Mixed practice - complete without looking back at the lessons.

1. Markup, Markdown and Profit	Cost=80, sell=104. Find profit.
2. Break-Even Analysis	Find contribution if $p=75$, $v=45$.
3. Taxes, Discounts and Consumer Finance	Find final price of 250 plus 20% tax.
4. Probability and Financial Risk	Find EV of 0.6 chance 100 and 0.4 chance -20.
5. Financial Planning and Integrated Review	What should you identify first in a time-value problem?
6. Markup, Markdown and Profit	Find markup on cost.
7. Break-Even Analysis	Find break-even units if $F=12000$.
8. Taxes, Discounts and Consumer Finance	Find pre-tax price if final is 360 inclusive at 20%.
9. Probability and Financial Risk	Do probabilities in a complete outcome set sum to what?
10. Financial Planning and Integrated Review	Why compare effective rates?
11. Markup, Markdown and Profit	Find profit margin.
12. Break-Even Analysis	Find profit at $Q=500$ using same data.

Written Task

Explain one financial decision using at least four concepts from this test. Include at least one formula and one interpretation sentence.

Self-Assessment

- Score 10-12: Strong mastery.
- Score 7-9: Good progress; review minor errors.
- Score 4-6: Revisit the related lessons.
- Score 0-3: Study the lesson explanations before retrying.

Working Space

Use this section after completing the revision tests independently.

1. Equal regular payments over time.	2. $FV = R \left[\frac{(1+i)^n - 1}{i} \right]$.	3. $PV = R \left[1 - \frac{(1+i)^{-n}}{i} \right]$.
4. Time 0.	5. 2000.	6. Payments at end of periods.
7. 20.	8. PV of one unit paid each period.	9. Multiply by $1+i$.
10. 5000.	11. Payments at beginning of periods.	12. Periodic payment.

1. R=PV/annuity PV factor.	2. 40.	3. Regular deposits accumulated for a future obligation.
4. Opening balance x periodic rate.	5. 260.	6. R=FV/FV annuity factor.
7. Payment - interest.	8. 1740.	9. R falls.
10. Opening balance - principal repaid.	11. 1740.	12. R falls, other things equal.

- For every wrong answer, rewrite the calculation with the correct formula and units.
- If the error is repeated, return to the lesson and complete the Mastery Challenge again.

Use this section after completing the revision tests independently.

1. 2000/year.	2. 561.80.	3. 14%.
4. 11000.	5. 4.854% approximately.	6. 10%.
7. 6141.25.	8. It falls.	9. 4%.
10. Straight-line.	11. About 6%.	12. 10%.

1. Sum of discounted cash flows including time 0.	2. Rate making NPV zero.	3. 80.
4. 5%.	5. No.	6. 20%.
7. PV of coupons and PV of redemption value.	8. 10%.	9. PV of benefits exceeds costs at the required return.
10. Zero.	11. Price falls.	12. 15%.

- For every wrong answer, rewrite the calculation with the correct formula and units.
- If the error is repeated, return to the lesson and complete the Mastery Challenge again.

REFERENCE SKILLS

Algebra and Calculator Skills

Essential techniques for financial mathematics.

Rearranging Formulas

- From $I=Prt$, $r=I/(Pt)$, $t=I/(Pr)$, $P=I/(rt)$.
- From $A=P(1+r)^n$, growth factor is A/P and logarithms can solve for n when required.
- Keep percentages as decimals inside formulas unless a formula explicitly states otherwise.

Logarithms for Time

Concept / Formula	Explanation	Worked Example
Solve $A=P(1+r)^n$	$n = \ln(A/P) / \ln(1+r)$	Example: doubling at 8% gives $n=\ln 2/\ln 1.08$.
Solve compound rate	$r = (A/P)^{(1/n)} - 1$	Use when P , A and n are known.
CAGR	$(\text{Ending/Beginning})^{(1/n)} - 1$	Annual compound growth over n years.

Calculator Checklist

- Use brackets around full numerators and denominators.
- Store full precision during working; round the final result.
- Check whether calculator is in the correct mode for percentages/logarithms.
- Estimate roughly before trusting a decimal answer.

Practice

1. Rearrange $A=P(1+r)^n$ to make P the subject.
2. Rearrange $I=Prt$ to make t the subject.
3. Write the formula to solve r from $A=P(1+r)^n$.
4. Explain why early rounding can change long-term compound results.

Answers

- $P=A/(1+r)^n$.
- $t=I/(Pr)$.
- $r=(A/P)^{(1/n)} - 1$.
- Repeated calculations magnify rounding error over many periods.

How to present financial mathematics clearly.

Concept / Formula	Explanation	Worked Example
Calculate	Show method and obtain a numerical answer.	
Determine	Find the required value, usually from given data.	
Compare	State relevant similarities/differences using numbers when possible.	
Explain	Give a mathematical/financial reason, not only a result.	
Interpret	Describe what the number means in context.	
Evaluate	Discuss assumptions, limitations and implications before a supported conclusion.	

- Using 8 instead of 0.08 in a formula.
- Using years with a monthly rate or months with an annual rate.
- Using nominal and effective rates as if they were identical.
- Forgetting whether payments are beginning or end of period.
- Rounding too early.
- Reporting a number without currency, percentage or time-period context.

1. Define variables and rate period.
2. State formula.
3. Substitute with consistent units.
4. Calculate with full precision.
5. Round appropriately and interpret.

REFERENCE

Financial Mathematics Formula Sheet

Core formulas used throughout this course.

Concept / Formula	Explanation	Worked Example
Percentage change	$(\text{new-old})/\text{old} \times 100\%$	Use original value in denominator.
Simple interest	$I = Prt$; $A = P(1+rt)$	r as decimal; t in rate periods.
Compound amount	$A = P(1+r)^n$	r periodic rate.
Nominal compounding	$A = P(1+j/m)^{mt}$	j nominal annual rate.
Effective annual rate	$EAR = (1+j/m)^m - 1$	Use for comparison.
Present value	$PV = FV/(1+r)^n$	Reverse compounding.
Ordinary annuity FV	$FV = R[(1+i)^n - 1]/i$	End-period payments.
Ordinary annuity PV	$PV = R[1 - (1+i)^{-n}]/i$	End-period payments.
Annuity due	Ordinary value $\times (1+i)$	Beginning-period payments.
Perpetuity	$PV = R/i$	First payment in one period.
Growing perpetuity	$PV = R1/(i-g)$	Requires $i > g$.
Loan interest	Opening balance $\times i$	Periodic rate.
Straight-line depreciation	$(\text{Cost} - \text{Salvage})/\text{Life}$	Equal annual amount.
Reducing balance	$BV_n = \text{Cost}(1-r)^n$	Percentage on book value.
Exact real return	$(1+\text{nominal})/(1+\text{inflation}) - 1$	Purchasing-power return.
NPV	$\sum CF_t/(1+r)^t$	Include CF_0 .
Break-even quantity	$F/(p-v)$	Requires $p > v$.
Expected value	$\sum p_i \times x_i$	Probabilities sum to 1.

REFERENCE

Financial Mathematics Quick Guide

A final one-page checklist before an examination.

Percentages

- Find the correct base.
- Use multipliers for successive changes.
- Margin and markup use different denominators.

Time Value of Money

- Match rate and time period.
- Compounding moves money forward; discounting moves it backward.
- Draw a timeline for multiple cash flows.

Annuities & Loans

- Identify ordinary vs due.
- Use periodic i and correct n .
- Loan payment splits into interest and principal.

Investments

- Total return includes income and capital gain/loss.
- NPV is money; IRR is a rate.
- Bond price and yield move inversely.

Consumer & Business

- Discount, tax, payroll and markup depend on the stated base.
- Break-even uses contribution $p-v$.
- Inflation reduces purchasing power.

Final Check

- Rate decimal correct?
- Periods matched?
- Formula appropriate?
- Units shown?
- Rounding sensible?
- Answer interpreted in context?

Exam Reminder

- Show method, not only calculator output.
- Use full precision during working.
- Check whether a result should be a currency, percentage, number of periods or unit quantity.
- A reasonable estimate is one of the best ways to catch mistakes.